Rhyme scheme pattern detector

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I Abstract

Stanzas of lyrics typically have rhyming patterns associated with them. For example, if there is a stanza that contains four verses where the first and third share a similar rhyme scheme and the second and fourth share a similar rhyme scheme, then the verses follow an ABAB pattern. These patterns are often determined strictly by hand. In this paper, we attempt to create automated methods for determining these patterns. We approach this process by looking at the phonetics within any given verse and comparing them to those of the other verses. The three methods we came up with are comparing the rhyming phonetics of the last words, the rhyming phonetics of all the words, and all the phonetics of all words between verses. We apply these methods to poetics lyrics that have predetermined rhyming schemes to see how our methods compare.

II Introduction

II.a Background

Analyzing patterns in a stanza of verses may contribute as a learning element in a lyrical analyzer. The following methods of pattern finding could lead to a better understanding of poetic writing. Given a stanza, one can compare the phonetics of the verses against one another to determine a pattern. See the examples below.

Example. *Phonetics:*
xerox = Z IH1 R AA0 K S

Example. *Rhyme scheme:*
Input: *stanza = [$verse_0$, $verse_1$, $verse_2$, $verse_3$]*
If $verse_0$ and $verse_2$ share a pattern and $verse_1$ and $verse_3$ share a different pattern. Then, we receive the following
Output: *ABAB*

We explore three methods, Last Words, Threshold with Rhyme, and Threshold without Rhyme, for determining patterns in both song lyrics and Dr. Seuss books.

II.b Data

The data was scraped from the Spotify and Genius. From there, the actual pattern classification element was computed by hand and pasted into the correct field. The headers of our dataset are displayed below.

CSV Headers = [name, lyric, actual_pattern]

When working with the data, we utilized the Pronouncing Python API for converting words to phonetics and determining the rhyme portion of a word.

III Methods

Before getting into the specifics of each method, there are a few constants to keep in mind. The methods take verses one at a time and compare them with all future verses in the stanza. When comparing how verses match to one another, the methods will always choose exactly one verse to match with. This does not apply when there is only one verse in a stanza.

III.a Last Words Only

Our most basic method for determining patterns is Last Words. Last Words will remove everything except for the last word in each verse. Then, the program will determine the rhyming part of the last words and compare them.

Example. *Let's take*

$verse_1$ = I woke with a fright
$verse_2$ = In the middle of the night.
First, we remove the entire verse except for the last word: ***fright, night***
Now, we convert both words to their respective rhyming phonetic:
fright = AY1 T
night = AY1 T
Since these are the same, the verses are predicted to have an ***AA*** *pattern.*

III.b Threshold

Patterns are not always determined by the last word. For our next two methods, we compute a threshold of analysis. When comparing $verse_x$ and $verse_y$, the analysis looks at the position of a phonetic in $verse_x$ and compares it to a range of phonetics of $verse_y$.

Example. *In this example, we are determining what range of values in $verse_y$ to compare w to.*

$verse_x = [v, ..., w, ..., z]$
$verse_y = [a, ..., b]$

Both $verse_x$ and $verse_y$ are populated with phonetics.
Let's take $l_1 = length(verse_x)$ and $l_2 = length(verse_y)$.
We now find the percentile that w is in: $percentile = index(w)/l_1$.
To find our bounds, we add and subtract an arbitrary value from our percentile. In our case, we chose 0.1. If the calculated value goes outside of the range [0, 1], then the bound will be set equal to 0 or 1.

$lowerPercentile = percentile - 0.1$ *or* 0
$upperPercentile = percentile + 0.1$ *or* 1

We multiply both the lower and upper percentiles by l_2 to find what indexes to use.
Finally, we floor the lower bound and ceiling the upper bound to receive an integer value.

$lowerBound = floor(lowerPercentile * l_2)$
$upperBound = ceiling(upperPercentile * l_2)$

Our $verse_y$ can now be seen as

$verse_y = [a, ..., lowerBound : upperBound, ..., b]$

Thus, we compare w to all phonetics between the $lowerBound$ and $upperBound$ inclusive. This technique works for comparing a shorter verse to longer verse and vice versa.

Next, we compare only the rhyming phonetics of the words across verses. We refer to this method as Threshold with Rhyme. An issue arises when verse patterns are not determined by only the rhyming pattern, but possibly the alliteration from the beginning of the words. To do this, we simply remove the rhyming aspect and compare all phonetics of the verse. We call this method Threshold without Rhyme.

Example. *Let's take our verses from the Last Words example and see how to use the threshold method.*
$verse_1$ = I woke with a fright
$verse_2$ = In the middle of the night.

We convert these to their respective phonetics (without rhyme, but using rhyme is very similar).
$phonetic_1$ = AY1 W OW1 K W IH1 DH AH0 F R AY1 T
$phonetic_2$ = IH0 N DH AH0 M IH1 D AH0 L AH1 V DH AH0 N AY1 T
$phonetic_1$ and $phonetic_2$ have 12 and 16 total phonetics respectively.

Let's look at the sixth phonetic in $phonetic_1$, ***IH1****.*
First, we compute the percentile. $\frac{6}{12} = 0.5$.
Next, we find our upper and lower percentiles by adding and subtracting 10%. $0.4, 0.6$.

We then use our percentiles multiplied by the length to find bounds on $phonetic_2$.
$lowerBound = floor(0.4 * 16) = 6$.
$upperBound = ceiling(0.6 * 16) = 10$.

We take the phonetics in positions $6 - 10$ *from $phonetic_2$ for comparison.*
$6 - 10$ = *IH1 D AH0 L AH1.*
Lastly, we compare ***IH1*** *to this range of phonetics for any similarities.*
We compare all phonetics in $verse_1$ to their respective ranges of phonetics in $verse_2$.

IV Results

For our analysis of methods, we compared our method outputs to the actual patterns that are provided in the dataset. This data included three songs by the Red Hot Chili Peppers and three stories by Dr. Seuss. We begin our analysis with the Last Words method.

Last Words

Name	% Predicted Correctly
By the Way by Red Hot Chili Peppers	38.60%
Californication by Red Hot Chili Peppers	64.86%
Can't Stop by Red Hot Chili Peppers	97.14%
Green Eggs and Ham [Excerpt] by Dr. Seuss	77.46%
How the Grinch Stole Christmas by Dr. Seuss	98.04%
Oh the Places You'll Go! [Excerpt] by Dr. Seuss	59.68%

Figure 1: Percent of verses correct when using **Last Words**.

In Figure 1, we see Last Words correctly predicted more than 50% of the patterns in most of our sample data. In our threshold methods (Figure 2, Figure 3) we see similar predicting capabilities as Last Words except for Can't Stop by Red Hot Chili Peppers. With this specific song, Last Words does an excellent job at determining the correct pattern scheme. Threshold with Rhyme predicts this song worse overall, but not as poorly as Threshold without Rhyme predicts this song. After looking deeper into the lyrics for Can't Stop, we notice the patterns are mostly determined by the rhyme on the last word of the verse. Since both threshold methods look at the entire verse rather than the ending, the patterns will not be predicted as precisely.

Example. *Snippet from song:*
"All on a spaceship persevering"
"Use my hands for everything but steering"
These lyrics are in an ***AA*** *pattern; however, the pattern is not obvious without the last word.*

Threshold with Rhyme

Name	% Predicted Correctly
By the Way by Red Hot Chili Peppers	33.33%
Californication by Red Hot Chili Peppers	62.16%
Can't Stop by Red Hot Chili Peppers	45.71%
Green Eggs and Ham [Excerpt] by Dr. Seuss	63.38%
How the Grinch Stole Christmas by Dr. Seuss	99.02%
Oh the Places You'll Go! [Excerpt] by Dr. Seuss	62.90%

Figure 2: Percent of verses correct when using **Threshold with Rhyme**.

Threshold without Rhyme

Name	% Predicted Correctly
By the Way by Red Hot Chili Peppers	26.32%
Californication by Red Hot Chili Peppers	59.46%
Can't Stop by Red Hot Chili Peppers	15.71%
Green Eggs and Ham [Excerpt] by Dr. Seuss	66.20%
How the Grinch Stole Christmas by Dr. Seuss	99.02%
Oh the Places You'll Go! [Excerpt] by Dr. Seuss	56.45%

Figure 3: Percent of verses correct when using **Threshold without Rhyme**.

V Discussion and Conclusion

These methods do not produce consistent results. Last Words, Threshold with Rhyme, and Threshold without Rhyme all produce similar results apart from a single piece of datum. In the future, we could add functionality to the methods to not always choose a verse to match with. Adding this functionality helps determine patterns that only have a single instance of a given verse, *AABCC*. Currently if three verses share the same pattern, only two of the verses will match. We could implement a function to determine more than one verse to match, *ABABA*. Beyond these extra functions, automated learning could be implemented for better predictive analysis. Results from this project will be used to further analyze lyrics.

VI References

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